Multi-peak place fields: methods

Detection of more than one firing peak inside a single place field, as computed by `ArenaSessionType_StatsComparison_v20.py` (multi-peak block, originally `PlaceCell_MultiPeak_KDE_v2.py`). Place cells and their fields come from `PlaceCell_Main_v21.py`. Parameter values are those set in the script on 10 Nov 2026 (output folder `MultiPeak_KDE_Pk75_Tr02_Sm035_Skaggs_adaptive_binned`).

0. Overview

The question. Does one place field contain two or more separate firing peaks, or only one? This is a question about the inside of a single field. It is not about whether a cell has several separate fields; that is counted elsewhere.

The idea in one paragraph. For each place field we take the firing rate of every spatial bin in the field and build a smooth, continuous "rate landscape" over the field with a weighted kernel density estimate (KDE). We then look for hills (local maxima) in this landscape. A hill is counted as a real, separate peak only if (i) it is tall enough compared with the field's highest hill and (ii) the landscape dips into a clear valley (trough) between it and every higher hill. A field with two or more counted peaks is called *multi-peak*. Multi-peak fields are cut into sub-fields along the valleys between the peaks (watershed), and the depth of the valleys, the heights of the peaks and the distances between them are measured.

```
Tracking (.csv, 30 Hz) + spikes (.ntt)
  |
  |- 1. Take the place cells and their fields from PlaceCell_Main
  |- 2. Rebuild the rate maps of each cell (raw, Gaussian-smoothed, Skaggs adaptive)
  |- 3. Per field: fit a rate-weighted Gaussian KDE on the field's bins
  |- 4. Find the local maxima of the KDE ("candidate peaks")
  |- 5. Keep the candidates that are tall enough AND separated by a real trough
  |- 6. >= 2 kept peaks  ->  multi-peak field; cut it into sub-fields (watershed)
  |- 7. Measure trough depth, peak heights, peak separation; per-cell summaries
  '- 8. Compare arenas / session types / animals (Kruskal-Wallis, chi-square)
```

1. Cells and fields that go in

Only cells classified as place cells (`place_cell == True` in the `Full` sheet of the PlaceCell_Main workbook) are analysed. Their place fields are taken unchanged from the `PlaceFields` sheet (the list of bins of each field, `bin_coords`); the multi-peak analysis does not re-detect fields. Cells that PlaceCell_Main overruled after field detection (area caps, below) are not included.

For reference, PlaceCell_Main isolates fields on the Gaussian-smoothed rate map λ_s (section 2.6) with the "threshold method". A bin b qualifies if

$$\lambda_s(b) \;\geq\; 0.20 \max_{b'} \lambda_s(b') \quad \text{and} \quad \lambda_s(b) \;>\; \bar{\lambda},$$

where $\bar{\lambda}$ is the cell's mean firing rate. Qualifying bins are grouped into 8-connected regions (bins touching at an edge or a corner). A region is a field if it spans at least 9 bins and contains at least 10 spikes (moving frames only). A cell is no longer a place cell if any single field covers more than 50% of the visited area, or all its fields together cover more than 60%.

2. Rebuilding the rate maps

The multi-peak code recomputes each cell's rate maps from the raw tracking and spike files, with exactly the same steps and settings as PlaceCell_Main, so that the KDE sees the same maps the fields were isolated on. Only the whole session is used (no first/second half split).

2.1 Tracking clean-up

1. Positions are read from the `*_cm.csv` file (time in µs, x and y in cm). Frames whose x equals the tracker's "lost" value (±1) are removed.
2. Frames that imply a step speed of 90 cm/s or more are removed; the remaining frames are sorted in time and shifted so that $\min x = \min y = 0$.
3. Jump removal is repeated: any frame reached from the previous good frame at more than 90 cm/s is marked bad, until no new bad frames appear. Bad frames are filled in by linear interpolation in time between good frames.
4. x and y are each smoothed with a Gaussian filter of $\sigma = 5$ samples (about 0.17 s at 30 Hz).

2.2 Frame durations and running speed

Each tracking frame k (time t_k, smoothed position $\mathbf{p}_k = (x_k, y_k)$) is given a duration, capped at two frame intervals so that tracking gaps do not add false occupancy:

$$\Delta t_1 = \tfrac{1}{30}\ \text{s}, \qquad \Delta t_k = \min\!\big(t_k - t_{k-1},\ \tfrac{2}{30}\ \text{s}\big) \quad (k \geq 2).$$

Running speed is the distance to the next frame divided by the time to the next frame:

$$v_k = \frac{\|\mathbf{p}_{k+1} - \mathbf{p}_k\|}{t_{k+1} - t_k}.$$

Frames with $v_k < 0.5$ cm/s are treated as immobile and dropped from all maps.

2.3 Spikes

Spikes of cluster 0 (unsorted) are discarded. Each spike is assigned to the tracking frame nearest in time. A spike is kept only if that frame is within 50 ms of the spike and is a moving frame.

2.4 Bins, occupancy, spike counts, visits

The arena is divided into square bins of 2 cm × 2 cm. A frame at (x, y) falls in bin $b = (\lfloor x/2 \rfloor, \lfloor y/2 \rfloor)$. For every bin:

$$T(b) = \sum_{k \in b} \Delta t_k \quad (\text{occupancy, s}), \qquad N(b) = \#\{\text{kept spikes in } b\}.$$

A *visit* starts every time the animal enters a bin from a different bin. A bin is **valid** if $T(b) > 0$ and it was visited at least 2 times. (A minimum-occupancy rule exists in the code but is switched off.) Invalid bins have no rate.

2.5 Raw rate map

$$\lambda_{\text{raw}}(b) = \frac{N(b)}{T(b)} \quad (\text{Hz, valid bins only}).$$

2.6 Gaussian-smoothed rate map

The raw map is smoothed with a 2-D Gaussian kernel G of $\sigma = 3$ cm (1.5 bins), cut off at 3σ and normalised to sum to 1. Invalid bins are excluded and the result is edge-corrected by dividing by the smoothed mask M (1 for valid bins, 0 otherwise):

$$\lambda_s(b) = \frac{\big(G * (M\,\lambda_{\text{raw}})\big)(b)}{\big(G * M\big)(b)}, \qquad G(\Delta) \propto \exp\!\left(-\frac{\|\Delta\|^2}{2\sigma^2}\right),$$

where $*$ is 2-D convolution. λ_s is the map the fields were isolated on, and is used for the plots, the sub-field peak and centre of mass (section 7.3) and the consistency check (section 2.8).

2.7 Skaggs adaptive-binned rate map: the map the KDE is fitted on

The KDE is weighted with the **Skaggs adaptive-binned** rate map λ_A. The idea of adaptive binning is that a well-sampled bin can use a small neighbourhood, while a poorly sampled bin (little time, few spikes) borrows data from a larger neighbourhood. Around every valid bin b we grow a disc of bins, from radius 0 (the bin itself) outwards, and stop at the first disc that holds "enough data". For a disc whose bins lie within ρ bins of b ($\rho^2 = 0, 1, 2, \ldots, 199$), let

$$T_D = \sum_{b' \in D} T(b'), \qquad N_D = \sum_{b' \in D} N(b'), \qquad r = (\rho + 0.5) \times 2 \text{ cm},$$

with only valid bins contributing. The disc grows until

$$r \cdot T_D \cdot \sqrt{N_D} \;\geq\; \alpha, \qquad \alpha = 200 \;\; \text{cm} \cdot \text{s} \cdot \text{spike}^{1/2}.$$

If no disc up to $\rho^2 = 199$ (radius of about 14 bins) meets the criterion, the largest disc is used. The rate of bin b is then

$$\lambda_A(b) = \begin{cases} N_D / T_D & \text{if } T_D \geq 0.5 \text{ s} \\ 0 & \text{otherwise.} \end{cases}$$

This is a rearranged form of the Skaggs adaptive-binning criterion, $r \geq \alpha / (T\sqrt{N})$: the radius needed shrinks as occupancy and spike count grow. The same $\alpha = 200$ was used in the PlaceCell_Main run that produced the input workbook. (The code also offers raw, Gaussian, occupancy-weighted smoothed and Yartsev adaptive maps as KDE input; only the Skaggs map was used here.)

2.8 Consistency check

For every field, the highest λ_s over the field's bins is compared with the field peak stored by PlaceCell_Main (`peak_fr_hz`). If they differ (by more than both 0.001 Hz and 0.1%), the rebuilt maps do not reproduce the original run, `ratemap_check = False`, and the field is excluded (section 8).

3. Circle track

On the circular track the place field is a stretch of track, so the analysis is done in one dimension, along the track angle.

1. **Ring centre.** A circle is fitted to the moving positions by algebraic (Kåsa) least squares:
$$\min_{a,b,c} \sum_k \left(x_k^2 + y_k^2 - a x_k - b y_k - c\right)^2, \qquad (c_x, c_y) = \left(\tfrac{a}{2}, \tfrac{b}{2}\right).$$
2. **Angle and radius** of each frame: $\theta_k = \operatorname{atan2}(y_k - c_y,\; x_k - c_x) \bmod 2\pi$, $r_k = \|\mathbf{p}_k - \mathbf{c}\|$.
3. **Angular bins** about 2 cm long at the median radius $\tilde{r}$:
$$n_\theta = \operatorname{round}\left(\frac{2\pi}{2 \text{ cm} / \tilde{r}}\right), \qquad \Delta\theta = \frac{2\pi}{n_\theta}.$$
4. Occupancy, spikes, visits, valid bins and the raw map are built as in 2-D. Gaussian smoothing and the adaptive map are 1-D and wrap around $\theta = 0/2\pi$. The adaptive "disc" becomes an interval of $\pm\rho$ bins along the track, $\rho = 0, 1, \ldots, \min(14, \lfloor (n_\theta - 1)/2 \rfloor)$, again with $r = (\rho + 0.5) \times 2$ cm.
5. **Track width** (used only for sub-field areas): $w = r_{99.5\%} - r_{0.5\%}$, the 0.5–99.5 percentile range of r_k.
6. The bins of a field are put in track order. A field that crosses the $0/2\pi$ seam is rotated so that it starts after its gap. The bins then get positions $u = 0, 1, \ldots, L - 1$ along the track (L = number of bins in the field).

Everything below is written for the 2-D case. For the Circle track, read "position u" for "bin centre $\mathbf{x}$", 1-D for 2-D, and "the left and right bins" for "the 8 surrounding bins".

4. The rate landscape of a field (weighted KDE)

4.1 Points and weights

For a field with n bins, bin i has its centre at $\mathbf{x}_i = (b_x + 0.5,\ b_y + 0.5)$, in bin units, and weight $r_i = \lambda_A(b_i)$, its adaptive-map rate in Hz. Only bins with $r_i > 0$ enter the KDE. The normalised weights are

$$\tilde{w}_i = \frac{r_i}{\sum_j r_j}, \qquad \sum_i \tilde{w}_i = 1.$$

If fewer than 5 bins of the field have $r_i > 0$, no KDE is fitted (`kde_note` says why) and the field is excluded from the comparison.

4.2 Bandwidth

The KDE is `scipy.stats.gaussian_kde` with weights. First the rate-weighted mean and covariance of the bin centres are computed:

$$\boldsymbol{\mu} = \sum_i \tilde{w}_i\, \mathbf{x}_i, \qquad \boldsymbol{\Sigma} = \frac{\sum_i \tilde{w}_i\, (\mathbf{x}_i - \boldsymbol{\mu})(\mathbf{x}_i - \boldsymbol{\mu})^{\mathsf{T}}}{1 - \sum_i \tilde{w}_i^{\,2}}.$$

The kernel covariance is the data covariance scaled by a fixed bandwidth factor h:

$$\mathbf{H} = h^2\, \boldsymbol{\Sigma}, \qquad h = 0.35.$$

So along each axis the kernel's standard deviation is 0.35 × the field's rate-weighted spread along that axis. The kernel therefore scales with the field: a large field gets a wider kernel than a small field. The kernel width in cm is stored per field (`kde_bw_cm`), as is the effective number of points, $n_{\text{eff}} = 1/\sum_i \tilde{w}_i^{\,2}$ (`kde_n_eff`).

> **Narrow (1-bin-wide) fields.** If all non-zero bins lie on a line, $\boldsymbol{\Sigma}$ is singular and a 2-D KDE cannot be fitted. Let $\ell_{\min} \le \ell_{\max}$ be the eigenvalues of $\boldsymbol{\Sigma}$ and $\mathbf{e}$ the eigenvector of $\ell_{\max}$ (the field's long axis). If $\ell_{\min} \le 10^{-6}\, \ell_{\max}$, every bin is projected onto that axis, $u_i = \mathbf{e}^{\mathsf{T}}\mathbf{x}_i$, and a 1-D KDE is fitted on u instead (`kde_axes` records the axis). If $\ell_{\max} \le 0$ (all non-zero bins at one location), no KDE is fitted.

4.3 The rate landscape

The weighted KDE is a probability density, a sum of Gaussian bumps centred on the bins:

$$\hat{f}(\mathbf{x}) = \sum_i \tilde{w}_i\, \varphi_{\mathbf{H}}(\mathbf{x} - \mathbf{x}_i), \qquad \varphi_{\mathbf{H}}(\mathbf{z}) = \frac{1}{(2\pi)^{d/2}\,|\mathbf{H}|^{1/2}} \exp\!\left(-\tfrac{1}{2}\,\mathbf{z}^{\mathsf{T}}\mathbf{H}^{-1}\mathbf{z}\right),$$

with $d = 2$ (or 1). Multiplying by the summed rate turns it into a rate landscape in Hz:

$$R(\mathbf{x}) \;=\; \Big(\sum_j r_j\Big)\, \hat{f}(\mathbf{x}) \;=\; \sum_i r_i\, \varphi_{\mathbf{H}}(\mathbf{x} - \mathbf{x}_i).$$

Each bin contributes a Gaussian bump whose volume equals its rate. Because each kernel integrates to 1 over an area measured in bins, R behaves like a smoothed rate map in Hz. R is evaluated at the centre of every field bin, and also at a one-bin **ring** of bins just outside the field (used in section 5).

5. Candidate peaks (local maxima)

Write $\mathcal{N}(i)$ for the 8 bins surrounding field bin i, including ring bins outside the field. Bin i is a local maximum if it is at least as high as all of them:

$$R(\mathbf{x}_i) \;\ge\; R(\mathbf{x}_j) \quad \text{for all } j \in \mathcal{N}(i).$$

- Including the ring means that a bin on the field's edge only counts as a peak if the landscape really falls off outside the field. A field cut out of the side of a larger hill does not get a false peak at its border.
- Local maxima are ranked from highest to lowest. Two neighbouring local maxima can only be exact ties; of each such pair only the higher-ranked one is kept.
- If no field bin is a local maximum (in a non-convex field, the KDE's top can lie just outside the field), the field's highest bin is used as its single peak (`peak_fallback = True`).

Call the highest local maximum p_1 and its height $R^* = R(p_1)$ (`kde_peak_hz`).

6. Which candidates count as peaks

The highest candidate p_1 always counts. Every other candidate p counts only if it passes both of the following tests.

6.1 Height test

$$R(p) \;\geq\; \beta\, R^*, \qquad \beta = 0.75.$$

The second peak must reach at least 75% of the field's highest peak (`peak_threshold_hz` = $0.75\, R^*$).

6.2 Trough (valley) test

To walk from p to any point higher than p, you must go down into a valley first. The **merge level** $m(p)$ is the height of the best (highest) pass on that walk. Over all paths γ of neighbouring field bins that start at p and end at any bin q with $R(q) > R(p)$:

$$m(p) \;=\; \max_{\gamma:\, p \to q,\; R(q) > R(p)} \; \min_{i \in \gamma} R(\mathbf{x}_i).$$

In words: of all ways to climb to higher ground, take the one that loses the least height; $m(p)$ is the lowest point on that way, the saddle. Paths only run through the field's own bins. The candidate counts if the landscape dips more than 20% below the candidate's own height before rising to anything higher:

$$\frac{m(p)}{R(p)} \;<\; 1 - \delta, \qquad \delta = 0.2.$$

The ratio $m(p)/R(p)$ is stored for every local maximum (`local_max_trough_rel`). Flat ridges and plateaus give $m(p) \approx R(p)$ and never count as a second peak.

How $m(p)$ is computed. $m(p)$ is found for all bins at once with a union-find sweep, a form of 0-dimensional topological persistence. The bins are visited from the highest R to the lowest. Each new bin joins the regions of its already-visited neighbours. When it connects two regions, the region with the lower top dies, and its top's merge level is set to the R of the bin that connected them. The field's highest bin never merges ($m = \infty$).

> **Worked example.** A field has two hills: 10 Hz and 8 Hz.
> • Valley between them at 6 Hz: height 8 ≥ 0.75 × 10 = 7.5 ✓; trough 6 / 8 = 0.75 < 0.8 ✓ → **2 peaks, multi-peak**.
> • Valley at 7 Hz: trough 7 / 8 = 0.875 ≥ 0.8 ✗ → the dip is too shallow, **1 peak**.
> • Second hill 7 Hz with a deep valley at 3 Hz: height 7 < 7.5 ✗ → **1 peak**.

6.3 Field classification

$$n_{\text{peaks}} = \#\{\text{counted peaks}\}, \qquad \text{multi_peak} = \big(n_{\text{peaks}} \geq 2\big).$$

7. Sub-fields and measures

7.1 Cutting the field into sub-fields (watershed)

Each counted peak p_k seeds sub-field k. A marker-based watershed (priority flood) then grows all sub-fields together, starting with the highest bins. Repeatedly, the highest bin already labelled is taken, and every unlabelled neighbour inside the field gets its label. Each bin thus joins the peak whose basin reaches it first while descending the landscape, and neighbouring sub-fields meet along the valley floors. Every field bin ends up in exactly one sub-field. Non-counted local maxima are absorbed into a neighbouring sub-field.

7.2 Measures of a multi-peak field

Saddle between two touching sub-fields a and b: the highest crossing point along their shared border,

$$s_{ab} = \max_{\substack{i \in a,\, j \in b \\ j \in \mathcal{N}(i)}} \min\big(R(\mathbf{x}_i), R(\mathbf{x}_j)\big).$$

Trough ratio of that border, relative to the lower of the two peaks:

$$\tau_{ab} = \frac{s_{ab}}{\min\big(R(p_a), R(p_b)\big)}.$$

Field-level measures compared between groups (multi-peak fields only):

Column	Formula	Meaning
`shallowest_trough_rel`	$\max_{(a,b)} \tau_{ab}$	Depth of the weakest split (closest to 1 = shallowest valley). Usually below 0.8 by construction, because the trough test of section 6.2 uses a related ratio.
`lowest_counted_peak_rel`	$\min_k R(p_k)/R^*$	Height of the weakest counted peak relative to the highest; ≥ 0.75 by construction.
`peak_separation_cm`	$\mathrm{mean}_{(a,b)}\, d_{ab}$	Mean distance between the peaks of touching sub-fields.

The distance between two peak bins is measured between bin centres, converted to cm:

$d_{ab} = 2\ \mathrm{cm} \times \|\mathbf{x}_{p_a} - \mathbf{x}_{p_b}\|$ (2-D), $d_{ab} = 2\ \mathrm{cm} \times |u_{p_a} - u_{p_b}|$ (Circle, arc length along the tra

All pairs (a, b) above are pairs of sub-fields that share a border. Per-pair values are also stored (`troughs_rel`, `peak_separations_cm`).

7.3 Sub-field table (descriptive, not compared)

For each sub-field k (set of bins S_k): KDE peak height $R(p_k)$ and $R(p_k)/R^*$; the highest λ_s and its bin; the rate-weighted centre of mass on the Gaussian map,

$$\mathbf{c}_k = \frac{\sum_{b \in S_k} \lambda_s(b)\, \mathbf{b}}{\sum_{b \in S_k} \lambda_s(b)} \qquad \Big(\text{Circle: } \theta_{c,k} = \operatorname{atan2}\big(\textstyle\sum \lambda_s \sin\theta_b,\ \sum \lambda_s \cos\theta_b\big)\Big);$$

its area, $|S_k| \times 4\ \mathrm{cm}^2$ (Circle: arc length $|S_k| \times 2$ cm × track width w); its share of the field area, $100\,|S_k|/n$; and its highest saddle to a neighbouring sub-field, in Hz and as a fraction of $R(p_k)$.

7.4 Per-cell measures

Computed only from the fields kept after the exclusions of section 8. A cell with 3 fields, one excluded, counts as a 2-field cell.

$$n_{\text{multi}} = \sum_{\text{fields}} \text{multi_peak}, \qquad \text{pct_multi_peak_fields} = 100 \times \frac{n_{\text{multi}}}{n_{\text{fields}}},$$

$$\text{has_multi_peak_field} = (n_{\text{multi}} \geq 1), \qquad \text{max_n_peaks} = \max_{\text{fields}} n_{\text{peaks}}.$$

8. Exclusions

- **No KDE**: fewer than 5 non-zero bins in the adaptive map, or all non-zero bins at one location (`EXCLUDE_NO_KDE_FIELDS = True`).
- **Rate map mismatch**: `ratemap_check = False`, section 2.8 (`EXCLUDE_RATEMAP_MISMATCH = True`).

Excluded fields are still written to the `Fields` sheet of `MultiPeak_KDE_Fields.xlsx`, but they are left out of every comparison and of the per-cell values.

9. Statistical comparison

Groups are compared at the levels used throughout the script: arena type, session type within arena, animal-controlled arena comparisons (only animals recorded in all compared arenas), per animal, and all cells pooled (descriptives only, no tests). Three tables are compared: *Cell* (one row per place cell), *Field* (one row per field) and *MultiField* (multi-peak fields only).

- **Continuous measures** (`n_multi_peak_fields`, `pct_multi_peak_fields`, `max_n_peaks`, `n_peaks`, and the three MultiField measures): Kruskal–Wallis omnibus test, followed by pairwise Mann–Whitney U tests.
- **Categorical measures** (% of place cells with ≥ 1 multi-peak field; % of place fields that are multi-peak): chi-square test of independence on the group × yes/no table (Yates-corrected for 2 × 2 tables), followed by pairwise chi-square tests.
- Pairwise p-values are Holm–Bonferroni corrected within each measure. With m ordered p-values $p_{(1)} \leq \cdots \leq p_{(m)}$:

$$p_{(i)}^{\text{Holm}} = \max_{j \leq i} \; \min\bigl(1,\; (m - j + 1)\, p_{(j)}\bigr).$$

- $\alpha = 0.05$.

> **Independence caveat.** Kruskal–Wallis and Mann–Whitney treat every row as independent. Several fields of one cell, and one cell recorded in several sessions or arenas of a day, are repeated measures. The script's alternative, `CONTINUOUS_TEST = 'LMM'` (nested linear mixed model on ranks with cell, day and animal effects), accounts for this but was not the setting used.

10. Parameter summary

Parameter	Value	Role
`PEAK_REL_HEIGHT_FRAC` (β)	0.75	Height test: candidate ≥ 75% of highest KDE peak
`MIN_TROUGH_DIP_FRAC` (δ)	0.2	Trough test: dip below 80% of the candidate's height
`KDE_BW_METHOD` (h)	0.35	Kernel SD = 0.35 × field's rate-weighted SD
`MIN_KDE_POINTS`	5	Minimum non-zero bins to fit a KDE
`KDE_RATEMAP_TYPE`	Skaggs_adaptive_binned	Rate map giving the KDE weights

`MP_SKAGGS_ALPHA` (α)	200 cm·s·spike$^{1/2}$	Adaptive-disc stopping criterion
`MP_ADAPTIVE_MAX_RSQ`	200 ($\rho^2 \leq 199$)	Largest adaptive disc, about 14 bins radius
`MP_ADAPTIVE_MIN_OCC_S`	0.5 s	Disc occupancy below this → rate 0
`MP_BIN_CM`	2 cm	Bin size
`MP_GAUSSIAN_SIGMA_CM`	3 cm	Gaussian rate-map smoothing
`MP_MIN_VISITS`	2	Valid bin: visited ≥ 2 times
`MP_USE_MIN_OCC_S`	False	Minimum-occupancy rule off
`MP_IMMOBILITY_SPEED_CMS`	0.5 cm/s	Slower frames dropped
`MP_POS_JUMP_THRESH_CMS`	90 cm/s	Tracking jump threshold
`MP_POS_SMOOTH_SIGMA_SMP`	5 samples	Position smoothing
`MP_MAX_GAP_US`	50 ms	Max spike-to-frame gap
`MP_FPS`	30 Hz	Tracking frame rate
`MP_RADIAL_RANGE_CLIP_PCTILE`	0.5%	Circle track width percentiles
Field isolation (PlaceCell_Main)	20% of peak, > mean rate, ≥ 9 bins, ≥ 10 spikes	Input fields

11. Points worth knowing when writing this up

- **Three different maps play three roles.** Fields are defined on the Gaussian map λ_s; the KDE is weighted by the adaptive map λ_A; and peaks, troughs and sub-field borders are found on the KDE landscape R.
- **The bandwidth is relative.** $h = 0.35$ scales with each field's own spread, so the smoothing in cm differs between fields (stored in `kde_bw_cm`). Larger h merges peaks; smaller h splits them. The number of peaks found depends on h, β and δ.
- **Zero-rate bins** inside a field (adaptive rate 0) add nothing to the KDE, but R is still evaluated there, so they can form valleys.
- **The trough test** uses the valley to *any* higher ground, not only to counted peaks, and is relative to the candidate's own height. The reported `shallowest_trough_rel` uses the saddle between the final sub-fields, relative to the lower of the two peaks.
- **Peak separation** is only measured between sub-fields that touch. On the Circle track it is the arc length along the track, assuming about 2 cm per bin.
- **Outputs.** `MultiPeak_KDE_Fields.xlsx` (sheets Fields, SubFields, Cells, Settings), and one figure per cell in `ratemaps_GaussianSmoothed\MultiPeak` or `\SinglePeak`. Each figure shows the trajectory with spikes, the Gaussian map with field outlines (black), trough cuts (white) and counted peaks (stars), the KDE input map, and the field KDE.